Simplex algorithm

Outline

The standard form of an LP problem in canonical max form

Basis of the LP problem in standard form and dictionaries
- Basic feasible solutions and reduced costs
- Finding an initial basic feasible solution

Pivoting operations
- Determining a new basic feasible solution
- Update of the dictionary

High-level pseudo-code of the simplex algorithm

The standard form of an LP problem in canonical max form

Introduction: The standard form of an LP problem in canonical max form

- The simplex algorithm works on LP problems in **standard maximization form**
- In this section we obtain the standard form from an LP problem in canonical maximization form, by adding one slack variable per constraint
- We describe the input data (the vectors $\boldsymbol{c}$, $\boldsymbol{b}$ and the matrix $\boldsymbol{A}$) and the identity matrix contained in $\boldsymbol{A}$
- We introduce the LP problem with two variables that we use as an example throughout the chapter

The LP problem in standard form

- We consider an LP problem in standard maximization form with $n + m$ non-negative variables and m equality linear constraints
- It is obtained from an LP problem in canonical maximization form with n non-negative variables and m less-than-or-equal-to linear constraints by adding m slack non-negative variables:

$$\max \quad \boldsymbol{c}' \boldsymbol{x}$$

$$\text{subject to} \quad \boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b},$$

$$\boldsymbol{x} \geq 0.$$

The input data

The input data of the LP problem are:

1. an objective-function column vector $\boldsymbol{c} \in \mathbb{Z}^{(n+m)}$ of $n+m$ rows;
2. a constraint matrix $\boldsymbol{A} \in \mathbb{Z}^{m \times (n+m)}$ of m rows and $n+m$ columns;
3. a right-hand side column vector $\boldsymbol{b} \in \mathbb{Z}^{m}$ of m rows.

The vectors $\boldsymbol{c}$, $\boldsymbol{b}$ and the matrix $\boldsymbol{A}$

$$\boldsymbol{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \\ 0 \\ \vdots \\ 0 \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & 1 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2n} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & 0 & 0 & \cdots & 1 \end{pmatrix}$$

The vector of the variables

- The $m + n$ non-negative variables can be written as a column vector $\boldsymbol{x}$ of $n + m$ rows:

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ x_{n+2} \\ \vdots \\ x_{n+m} \end{pmatrix}$$

The columns of the slack variables

- Since the LP problem is obtained by adding m slack variables to an LP problem in canonical form, we have:

$$\begin{pmatrix} a_{1(n+1)} & a_{1(n+2)} & \cdots & a_{1(n+m)} \\ a_{2(n+1)} & a_{2(n+2)} & \cdots & a_{2(n+m)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m(n+1)} & a_{m(n+2)} & \cdots & a_{m(n+m)} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

- This is the identity matrix $\boldsymbol{I}$ of size m

The costs of the slack variables

- For the same reason, we have:

$$\begin{pmatrix} c_{n+1} \\ c_{n+2} \\ \vdots \\ c_{n+m} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

- The slack variables do not contribute to the objective function

The rank of the constraint matrix

- Since the constraint matrix $\boldsymbol{A} \in \mathbb{Z}^{m \times (n+m)}$ contains the identity matrix $\boldsymbol{I}$ of size m, we have:

$$\text{rank}(\boldsymbol{A}) = m$$

- Accordingly there are m **linearly independent rows** and at least one set of m **linearly independent columns** in the constraint matrix $\boldsymbol{A}$
- Recall: the row rank and the column rank of a matrix are equal

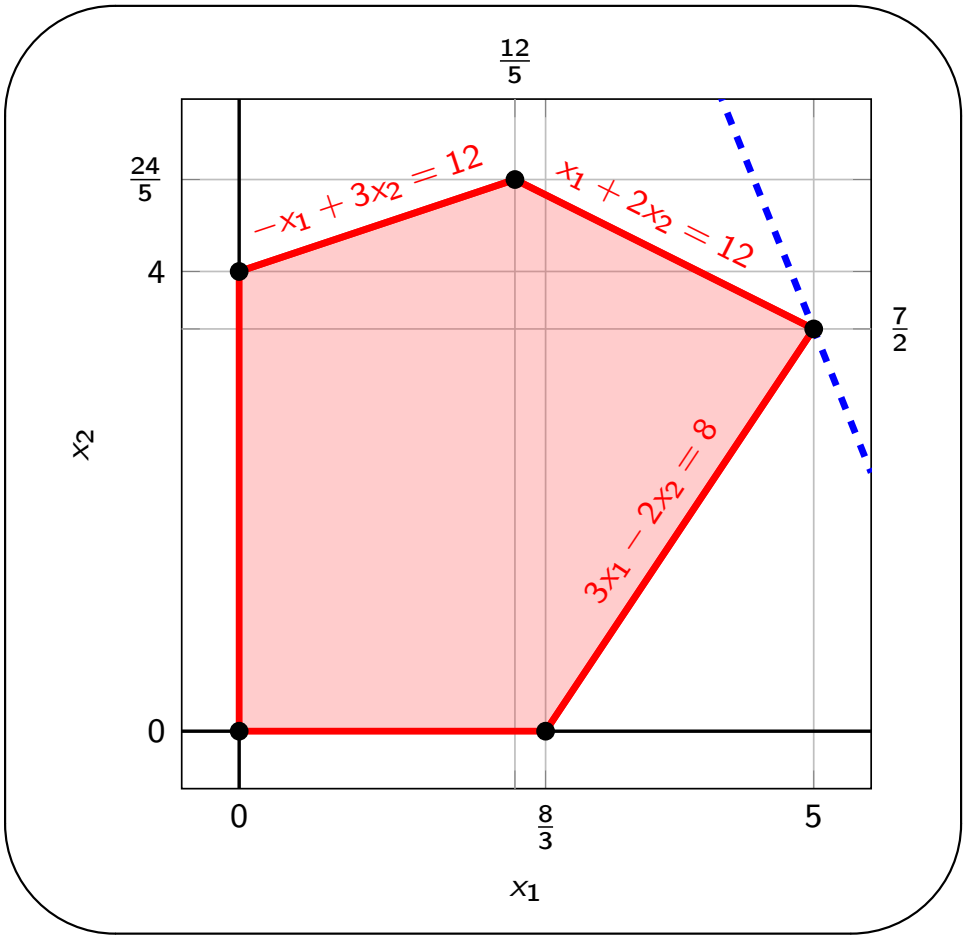
We now introduce the LP problem with two variables that we use as an example in the whole chapter.

Example (1/4)

- We consider the following LP problem in canonical max form with $n = 2$ variables and $m = 3$ constraints:

$$
\begin{array}{rrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\
 & & x_1 & + & 2 & x_2 & \leq & 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

Example (2/4)



In dashed blue the level set $5\,x_1 + 2\,x_2 = 32$ through the optimal vertex.

An optimal solution $\tilde{\boldsymbol{x}}(\,LP\,)$ is:

$$\tilde{x}_1 = 5, \quad \tilde{x}_2 = \frac{7}{2}$$

and the optimal value is:

$$z(\,LP\,) = 32$$

Example (3/4)

- The standard form with $n + m = 5$ variables and $m = 3$ constraints reads as follows:

$$
\begin{array}{rrrrrrrrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & & & & & & & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & + & x_3 & & & & & = & 12 \\
 & & x_1 & + & 2 & x_2 & & & + & x_4 & & & = & 12 \\
 & 3 & x_1 & - & 2 & x_2 & & & & & + & x_5 & = & 8 \\
 & & x_1, & & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0
\end{array}
$$

- We call it the **example problem**: we use it in all the examples of this chapter

Example (4/4)

- In the example problem in standard form, we have:

$$\boldsymbol{c} = \begin{pmatrix} 5 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix} -1 & 3 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 3 & -2 & 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}$$

- An optimal solution $\tilde{\boldsymbol{x}}(\, LP\,)$ of the standard form is:

$$\tilde{x}_1 = 5, \quad \tilde{x}_2 = \frac{7}{2}, \quad \tilde{x}_3 = \frac{13}{2}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0$$

Summary

- The standard form $\max\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq 0\}$ has $n + m$ variables and m equality constraints
- The columns of the m slack variables form the **identity matrix** $\boldsymbol{I}$, and their costs are zero
- Hence $\text{rank}(\boldsymbol{A}) = m$: there are m linearly independent columns in $\boldsymbol{A}$
- The example problem has $n = 2$, $m = 3$ and the optimal vertex $(5, \frac{7}{2})$ of value 32

Basis of the LP problem in standard form and dictionaries

Introduction: Basis of the LP problem in standard form and dictionaries

- We choose m linearly independent columns of $\boldsymbol{A}$: a **basis**
- A basis splits the variables into **basic** and **non-basic** variables
- Writing the basic variables and the objective function in function of the non-basic variables gives a **dictionary**: the object on which the simplex algorithm works
- We then define basic feasible solutions, reduced costs, and how to find an initial basic feasible solution

Definition 1: basis of the LP problem

Given a subset $\mathcal{B} \subseteq \{1, 2, \ldots, n+m\}$ *of* m *indices of the* $n+m$ *variables of the LP problem in standard form, if the matrix* $\boldsymbol{A}_{\mathcal{B}} \in \mathbb{R}^{m \times m}$ *containing the* m *columns of* $\boldsymbol{A}$ *with indices* $j \in \mathcal{B}$*:*

$$\boldsymbol{A}_{\mathcal{B}} = (\boldsymbol{A}_j)_{j \in \mathcal{B}}$$

is invertible, it is called a ***basis*** *of the LP problem in standard form.*

Definition 2: partition of the indices, basic and non-basic variables (1/2)

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, we have the following ***partition*** *of the indices $\{1, 2, \ldots, n+m\}$ of the variables:*

$$\mathcal{B} \subseteq \{1, 2, \ldots, n+m\}, \qquad |\mathcal{B}| = m,$$

$$\mathcal{N} = \{1, 2, \ldots, n+m\} \setminus \mathcal{B}, \qquad |\mathcal{N}| = n.$$

Definition: partition of the indices, basic and non-basic variables (2/2)

Given a partition $\mathcal{B}$ and $\mathcal{N}$, we define:

1. *the column vector of the **basic variables** containing the m variables with indices $i \in \mathcal{B}$: $\boldsymbol{x}_\mathcal{B} = (x_i)_{i \in \mathcal{B}}$*
2. *the column vector of the **non-basic variables** containing the n variables with indices $i \in \mathcal{N}$: $\boldsymbol{x}_\mathcal{N} = (x_i)_{i \in \mathcal{N}}$*
3. *the column vector $\boldsymbol{c}_\mathcal{B} \in \mathbb{R}^m$ containing the m values of $\boldsymbol{c}$ with indices $i \in \mathcal{B}$: $\boldsymbol{c}_\mathcal{B} = (c_i)_{i \in \mathcal{B}}$*
4. *the matrix $\boldsymbol{A}_\mathcal{N} \in \mathbb{R}^{m \times n}$ containing the n columns of $\boldsymbol{A}$ with indices $j \in \mathcal{N}$: $\boldsymbol{A}_\mathcal{N} = (\boldsymbol{A}_j)_{j \in \mathcal{N}}$*
5. *the column vector $\boldsymbol{c}_\mathcal{N} \in \mathbb{R}^n$ containing the n values of $\boldsymbol{c}$ with indices $i \in \mathcal{N}$: $\boldsymbol{c}_\mathcal{N} = (c_i)_{i \in \mathcal{N}}$*

Let us now check two choices of $\mathcal{B}$
for the example problem:

we test whether $\boldsymbol{A}_{\mathcal{B}}$ is invertible
and compute its inverse with the method of Gauss-Jordan.

Example: $\mathcal{B} = \{1, 3, 4\}$ (1/6)

- For the example problem with $\mathcal{B} = \{1, \ 3, \ 4\}$, we have:

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix}$$

$$\begin{aligned} \det(\boldsymbol{A}_{\mathcal{B}}) = & -1 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 3 + 0 \cdot 1 \cdot 0 \\ & - 1 \cdot 1 \cdot 0 - (-1) \cdot 1 \cdot 0 - 0 \cdot 0 \cdot 3 = 3 \neq 0 \end{aligned}$$

- Then $\text{rank}(\boldsymbol{A}_{\mathcal{B}}) = 3$: the matrix $\boldsymbol{A}_{\mathcal{B}}$ is invertible and it is a basis of the LP problem

Example: $\mathcal{B} = \{1, 3, 4\}$ (2/6)

- We compute the inverse matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$ with the method of Gauss-Jordan:

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 3 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 3 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$(R_1 \leftarrow -R_1)$$

Example: $\mathcal{B} = \{1, 3, 4\}$ (3/6)

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 3 & 0 & 3 & 0 & 1 \end{array}\right)$$

$$(R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - 3R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & -3 & 0 & -3 & 1 \end{array}\right)$$

$$(R_1 \leftarrow R_1 + R_2,\ R_3 \leftarrow R_3 - 3R_2)$$

Example: $\mathcal{B} = \{1, 3, 4\}$ (4/6)

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -\frac{1}{3} \end{array}\right)$$

$$(R_3 \leftarrow -\tfrac{1}{3} R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 1 & 0 & 1 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & 0 & 1 & -\frac{1}{3} \end{array}\right)$$

$$(R_1 \leftarrow R_1 - R_3,\ R_2 \leftarrow R_2 - R_3)$$

- The right half of the last matrix is $\boldsymbol{A}_{\mathcal{B}}^{-1}$

Example: $\mathcal{B} = \{1, 3, 4\}$ (5/6)

- Since $\mathcal{B} = \{1, \ 3, \ 4\}$, we have:

$$\boldsymbol{x}_{\mathcal{B}} = \begin{pmatrix} x_1 \\ x_3 \\ x_4 \end{pmatrix} \qquad \boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix} \qquad \boldsymbol{c}_{\mathcal{B}} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix}$$

$$\boldsymbol{A}_{\mathcal{B}}^{-1} = \begin{pmatrix} 0 & 0 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} \end{pmatrix}$$

Example: $\mathcal{B} = \{1, 3, 4\}$ (6/6)

- Accordingly $\mathcal{N} = \{2, \quad 5\}$ and we have:

$$\boldsymbol{x}_{\mathcal{N}} = \begin{pmatrix} x_2 \\ x_5 \end{pmatrix} \qquad \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} 3 & 0 \\ 2 & 0 \\ -2 & 1 \end{pmatrix} \qquad \boldsymbol{c}_{\mathcal{N}} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

Example: $\mathcal{B} = \{1, 2, 4\}$ (1/6)

- For the example problem with $\mathcal{B} = \{1, \ 2, \ 4\}$, we have:

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 3 & 0 \\ 1 & 2 & 1 \\ 3 & -2 & 0 \end{pmatrix}$$

and

$$\begin{aligned} \det(\boldsymbol{A}_{\mathcal{B}}) &= -1 \cdot (2 \cdot 0 - 1 \cdot (-2)) - 3 \cdot (1 \cdot 0 - 1 \cdot 3) \\ &\quad + 0 \cdot (1 \cdot (-2) - 2 \cdot 3) \\ &= -1 \cdot 2 + 3 \cdot 3 = 7 \neq 0 \end{aligned}$$

- Then $\text{rank}(\boldsymbol{A}_{\mathcal{B}}) = 3$: the matrix $\boldsymbol{A}_{\mathcal{B}}$ is invertible and it is a basis of the LP problem

Example: $\mathcal{B} = \{1, 2, 4\}$ (2/6)

- We compute the inverse matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$ with the method of Gauss-Jordan:

$$\left(\begin{array}{ccc|ccc} -1 & 3 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 3 & -2 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 3 & -2 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$(R_1 \leftarrow -R_1)$$

Example: $\mathcal{B} = \{1, 2, 4\}$ (3/6)

$$\left(\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ 0 & 5 & 1 & 1 & 1 & 0 \\ 0 & 7 & 0 & 3 & 0 & 1 \end{array}\right)$$

$$(R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - 3R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & -3 & 0 & -1 & 0 & 0 \\ 0 & 1 & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 7 & 0 & 3 & 0 & 1 \end{array}\right)$$

$$(R_2 \leftarrow \tfrac{1}{5} R_2)$$

Example: $\mathcal{B} = \{1, 2, 4\}$ (4/6)

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{5} & -\frac{2}{5} & \frac{3}{5} & 0 \\ 0 & 1 & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & -\frac{7}{5} & \frac{8}{5} & -\frac{7}{5} & 1 \end{array}\right)$$

$$(R_1 \leftarrow R_1 + 3R_2,\ R_3 \leftarrow R_3 - 7R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{5} & -\frac{2}{5} & \frac{3}{5} & 0 \\ 0 & 1 & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 1 & -\frac{8}{7} & 1 & -\frac{5}{7} \end{array}\right)$$

$$(R_3 \leftarrow -\tfrac{5}{7} R_3)$$

Example: $\mathcal{B} = \{1, 2, 4\}$ (5/6)

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{2}{7} & 0 & \frac{3}{7} \\ 0 & 1 & 0 & \frac{3}{7} & 0 & \frac{1}{7} \\ 0 & 0 & 1 & -\frac{8}{7} & 1 & -\frac{5}{7} \end{array}\right)$$

$$(R_1 \leftarrow R_1 - \tfrac{3}{5}R_3,\ R_2 \leftarrow R_2 - \tfrac{1}{5}R_3)$$

- Since $\mathcal{B} = \{1, \ 2, \ 4\}$, we have:

$$\boldsymbol{x}_{\mathcal{B}} = \begin{pmatrix} x_1 \\ x_2 \\ x_4 \end{pmatrix} \quad \boldsymbol{c}_{\mathcal{B}} = \begin{pmatrix} 5 \\ 2 \\ 0 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{B}}^{-1} = \begin{pmatrix} \frac{2}{7} & 0 & \frac{3}{7} \\ \frac{3}{7} & 0 & \frac{1}{7} \\ -\frac{8}{7} & 1 & -\frac{5}{7} \end{pmatrix}$$

Example: $\mathcal{B} = \{1, 2, 4\}$ (6/6)

- Accordingly $\mathcal{N} = \{3, \quad 5\}$ and we have:

$$\boldsymbol{x}_{\mathcal{N}} = \begin{pmatrix} x_3 \\ \\ x_5 \end{pmatrix} \qquad \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \qquad \boldsymbol{c}_{\mathcal{N}} = \begin{pmatrix} 0 \\ \\ 0 \end{pmatrix}$$

With a basis we can now rewrite the constraints
and the objective function
in function of the non-basic variables only.

Rewriting the equality constraints

- Given a basis $\boldsymbol{A}_{\mathcal{B}}$, we can re-write the equality constraints as follows:

$$\boldsymbol{A}\,\boldsymbol{x} = \begin{pmatrix} \boldsymbol{A}_{\mathcal{B}} & \boldsymbol{A}_{\mathcal{N}} \end{pmatrix} \begin{pmatrix} \boldsymbol{x}_{\mathcal{B}} \\ \boldsymbol{x}_{\mathcal{N}} \end{pmatrix} = \boldsymbol{A}_{\mathcal{B}}\,\boldsymbol{x}_{\mathcal{B}} + \boldsymbol{A}_{\mathcal{N}}\,\boldsymbol{x}_{\mathcal{N}} = \boldsymbol{b}$$

- Then, by multiplying by $\boldsymbol{A}_{\mathcal{B}}^{-1}$, we have:

$$\begin{aligned} & \boldsymbol{A}_{\mathcal{B}}^{-1}\left(\boldsymbol{A}_{\mathcal{B}}\,\boldsymbol{x}_{\mathcal{B}} + \boldsymbol{A}_{\mathcal{N}}\,\boldsymbol{x}_{\mathcal{N}}\right) = \boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{b} \\ \Rightarrow \quad & \underbrace{\boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{A}_{\mathcal{B}}}_{=\boldsymbol{I}}\,\boldsymbol{x}_{\mathcal{B}} + \boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{A}_{\mathcal{N}}\,\boldsymbol{x}_{\mathcal{N}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{b} \end{aligned}$$

The basic variables in function of the non-basic variables

The basic variables can be written in function of the non-basic variables as follows:

$$\boldsymbol{x}_{\mathcal{B}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} - \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}}\, \boldsymbol{x}_{\mathcal{N}}$$

Rewriting the objective function

- We can re-write the objective function value, denoted by z, as follows:

$$z = \boldsymbol{c}'\boldsymbol{x} = \begin{pmatrix} \boldsymbol{c}_{\mathcal{B}} \\ \boldsymbol{c}_{\mathcal{N}} \end{pmatrix}' \begin{pmatrix} \boldsymbol{x}_{\mathcal{B}} \\ \boldsymbol{x}_{\mathcal{N}} \end{pmatrix} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{x}_{\mathcal{B}} + \boldsymbol{c}'_{\mathcal{N}}\, \boldsymbol{x}_{\mathcal{N}}$$

- Using the expression of $\boldsymbol{x}_{\mathcal{B}}$ in function of $\boldsymbol{x}_{\mathcal{N}}$, we then get:

$$z = \boldsymbol{c}'_{\mathcal{B}} \left(\boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} - \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}}\, \boldsymbol{x}_{\mathcal{N}}\right) + \boldsymbol{c}'_{\mathcal{N}}\, \boldsymbol{x}_{\mathcal{N}}$$

The objective function in function of the non-basic variables

The objective function can be written in function of the non-basic variables as follows:

$$z = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}\ \boldsymbol{b} + \left(\boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}\ \boldsymbol{A}_{\mathcal{N}}\right) \boldsymbol{x}_{\mathcal{N}}$$

- With these two expressions, the **LP problem in standard form can be reformulated in form of a dictionary**

Definition 3: dictionary

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, the associated ***dictionary*** *is a system of $m+1$ linear equations in which the basic variables and the objective function are written in function of the non-basic variables:*

$$\begin{cases} \boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \bar{\boldsymbol{A}}\, \boldsymbol{x}_{\mathcal{N}} \\ \\ z = \bar{z} + \bar{\boldsymbol{c}}'\, \boldsymbol{x}_{\mathcal{N}} \end{cases}$$

where

$$\begin{cases} \bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} \\ \bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}} \\ \bar{z} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} \\ \bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}} \end{cases}$$

The elements of a dictionary

- Given a basis $\boldsymbol{A}_{\mathcal{B}}$, in the associated dictionary there are:
  1. the matrix $\bar{\boldsymbol{A}} \in \mathbb{R}^{m \times n}$ of m rows associated to the basic variables and n columns associated to the non-basic variables
  2. two column vectors $\bar{\boldsymbol{b}} \in \mathbb{R}^m$ and $\bar{\boldsymbol{c}} \in \mathbb{R}^n$ of m rows associated to the basic variables and of n rows associated to the non-basic variables, respectively:

$$\bar{\boldsymbol{A}} = (\bar{a}_{ij})_{i \in \mathcal{B}, j \in \mathcal{N}}, \qquad \bar{\boldsymbol{b}} = (\bar{b}_i)_{i \in \mathcal{B}}, \qquad \bar{\boldsymbol{c}} = (\bar{c}_i)_{i \in \mathcal{N}}.$$

A dictionary with the summations

A dictionary can be equivalently rewritten using the summations as follows:

$$\begin{cases} x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij}\, x_j, \quad \forall i \in \mathcal{B} \\ z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j\, x_j \end{cases}$$

Let us compute the dictionaries
associated with the two bases of the example problem
that we studied above.

Example: dictionary with $\mathcal{B} = \{1, 3, 4\}$ (1/6)

- For the example problem, with $\mathcal{B} = \{1, \ 3, \ 4\}$ and $\mathcal{N} = \{2, \ 5\}$, we have:

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} = \begin{pmatrix} 0 & 0 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} = \begin{pmatrix} \frac{8}{3} \\ \frac{44}{3} \\ \frac{28}{3} \end{pmatrix}$$

$$\bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} 0 & 0 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 2 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} & \frac{1}{3} \\ \frac{7}{3} & \frac{1}{3} \\ \frac{8}{3} & -\frac{1}{3} \end{pmatrix}$$

Example: dictionary with $\mathcal{B} = \{1, 3, 4\}$ (2/6)

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} = \begin{pmatrix} 5 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & \frac{5}{3} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} = \frac{40}{3}$$

Example: dictionary with $\mathcal{B} = \{1, 3, 4\}$ (3/6)

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_{\mathcal{N}}$$

$$= \begin{pmatrix} 2 & 0 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 2 & 0 \\ -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & \frac{5}{3} \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 2 & 0 \\ -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 \end{pmatrix} - \begin{pmatrix} -\frac{10}{3} & \frac{5}{3} \end{pmatrix} = \begin{pmatrix} \frac{16}{3} & -\frac{5}{3} \end{pmatrix}$$

Example: dictionary with $\mathcal{B} = \{1, 3, 4\}$ (4/6)

- Accordingly, the associated dictionary is:

$$\begin{cases} \begin{pmatrix} x_1 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} \frac{8}{3} \\ \frac{44}{3} \\ \frac{28}{3} \end{pmatrix} - \begin{pmatrix} -\frac{2}{3} & \frac{1}{3} \\ \frac{7}{3} & \frac{1}{3} \\ \frac{8}{3} & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} x_2 \\ x_5 \end{pmatrix} \\ z = \frac{40}{3} + \begin{pmatrix} \frac{16}{3} & -\frac{5}{3} \end{pmatrix} \begin{pmatrix} x_2 \\ x_5 \end{pmatrix} \end{cases}$$

Example: dictionary with $\mathcal{B} = \{1, 3, 4\}$ (5/6)

- or, equivalently:

$$\begin{cases} x_1 = \frac{8}{3} + \frac{2}{3} x_2 - \frac{1}{3} x_5 \\ x_3 = \frac{44}{3} - \frac{7}{3} x_2 - \frac{1}{3} x_5 \\ x_4 = \frac{28}{3} - \frac{8}{3} x_2 + \frac{1}{3} x_5 \\ z = \frac{40}{3} + \frac{16}{3} x_2 - \frac{5}{3} x_5 \end{cases}$$

Example: dictionary with $\mathcal{B} = \{1, 3, 4\}$ (6/6)

- This dictionary is associated to the **feasible** solution

$$\bar{x}_1 = \frac{8}{3}, \quad \bar{x}_2 = 0, \quad \bar{x}_3 = \frac{44}{3}, \quad \bar{x}_4 = \frac{28}{3}, \quad \bar{x}_5 = 0$$

  of value $z = \frac{40}{3}$

- All the basic variables take non-negative values

Example: dictionary with $\mathcal{B} = \{1, 2, 4\}$ (1/6)

- For the example problem, with $\mathcal{B} = \{1, \ 2, \ 4\}$ and $\mathcal{N} = \{3, \ 5\}$, we have:

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} = \begin{pmatrix} \frac{2}{7} & 0 & \frac{3}{7} \\ \frac{3}{7} & 0 & \frac{1}{7} \\ -\frac{8}{7} & 1 & -\frac{5}{7} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} = \begin{pmatrix} \frac{48}{7} \\ \frac{44}{7} \\ -\frac{52}{7} \end{pmatrix}$$

$$\bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} \frac{2}{7} & 0 & \frac{3}{7} \\ \frac{3}{7} & 0 & \frac{1}{7} \\ -\frac{8}{7} & 1 & -\frac{5}{7} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{7} & \frac{3}{7} \\ \frac{3}{7} & \frac{1}{7} \\ -\frac{8}{7} & -\frac{5}{7} \end{pmatrix}$$

Example: dictionary with $\mathcal{B} = \{1, 2, 4\}$ (2/6)

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} = \begin{pmatrix} 5 & 2 & 0 \end{pmatrix} \begin{pmatrix} \frac{2}{7} & 0 & \frac{3}{7} \\ \frac{3}{7} & 0 & \frac{1}{7} \\ -\frac{8}{7} & 1 & -\frac{5}{7} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{16}{7} & 0 & \frac{17}{7} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} = \frac{328}{7}$$

Example: dictionary with $\mathcal{B} = \{1, 2, 4\}$ (3/6)

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}}$$

$$= \begin{pmatrix} 0 & 0 \end{pmatrix} - \begin{pmatrix} 5 & 2 & 0 \end{pmatrix} \begin{pmatrix} \frac{2}{7} & 0 & \frac{3}{7} \\ \frac{3}{7} & 0 & \frac{1}{7} \\ -\frac{8}{7} & 1 & -\frac{5}{7} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \end{pmatrix} - \begin{pmatrix} \frac{16}{7} & 0 & \frac{17}{7} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \end{pmatrix} - \begin{pmatrix} \frac{16}{7} & \frac{17}{7} \end{pmatrix} = \begin{pmatrix} -\frac{16}{7} & -\frac{17}{7} \end{pmatrix}$$

Example: dictionary with $\mathcal{B} = \{1, 2, 4\}$ (4/6)

- Accordingly, the associated dictionary is:

$$\begin{cases} \begin{pmatrix} x_1 \\ x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} \frac{48}{7} \\ \frac{44}{7} \\ -\frac{52}{7} \end{pmatrix} - \begin{pmatrix} \frac{2}{7} & \frac{3}{7} \\ \frac{3}{7} & \frac{1}{7} \\ -\frac{8}{7} & -\frac{5}{7} \end{pmatrix} \begin{pmatrix} x_3 \\ x_5 \end{pmatrix} \\ z = \frac{328}{7} + \begin{pmatrix} -\frac{16}{7} & -\frac{17}{7} \end{pmatrix} \begin{pmatrix} x_3 \\ x_5 \end{pmatrix} \end{cases}$$

Example: dictionary with $\mathcal{B} = \{1, 2, 4\}$ (5/6)

- or, equivalently:

$$\begin{cases} x_1 = \frac{48}{7} - \frac{2}{7} x_3 - \frac{3}{7} x_5 \\ x_2 = \frac{44}{7} - \frac{3}{7} x_3 - \frac{1}{7} x_5 \\ x_4 = -\frac{52}{7} + \frac{8}{7} x_3 + \frac{5}{7} x_5 \\ z = \frac{328}{7} - \frac{16}{7} x_3 - \frac{17}{7} x_5 \end{cases}$$

Example: dictionary with $\mathcal{B} = \{1, 2, 4\}$ (6/6)

- This dictionary is **not** associated to a feasible solution
- Indeed, with $x_3 = x_5 = 0$ the basic variable x_4 takes the value $-\frac{52}{7} < 0$

Dependent and independent variables

- The basic variable values depend on the non-basic variable values, accordingly:

1. **the basic variables are the dependent variables**
2. **the non-basic variables are the independent variables**

Definition 4: basic solution

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, by setting all the non-basic variables to zero we have:

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \qquad \text{and} \qquad \boldsymbol{x}_{\mathcal{N}} = 0$$

*which is called a **basic solution** of the LP problem in standard form.*

- The variables are non-negative, so not all basic solutions are feasible solutions

Definition 5: basic feasible solutions

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, if $\bar{\boldsymbol{b}} \geq 0$ then

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \quad \text{and} \quad \boldsymbol{x}_{\mathcal{N}} = 0$$

*is called a **basic feasible solution**.*

- In the examples above, the basis with $\mathcal{B} = \{1, 3, 4\}$ gives a basic feasible solution, the basis with $\mathcal{B} = \{1, 2, 4\}$ does not

Definition 6: reduced cost vectors

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, the vector $\bar{\boldsymbol{c}}$ is called the ***reduced costs vector****.*

The meaning of a reduced cost

- The reduced cost

$$\bar{c}_p = c_p - \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_p$$

  of a non-basic variable x_p with $p \in \mathcal{N}$ is the change of the objective function per unit increment of the variable

- The increment is proportional to c_p
- The decrement (due to the change of the basic variables to preserve the equality constraints) is proportional to $\boldsymbol{c}'_{\mathcal{B}} \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_p$

To start, the simplex algorithm needs
a first basic feasible solution:

when $\boldsymbol{b} \geq 0$ the slack variables give one for free.

The initial basic feasible solution

- For an LP problem in standard form, if $\boldsymbol{b} \geq 0$, the partition:

$$\mathcal{B} = \{n+1, n+2, \ldots, n+m\} \quad \text{and} \quad \mathcal{N} = \{1, 2, \ldots, n\}$$

is associated to a basis $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$ and to the **initial basic feasible solution**:

$$\boldsymbol{x}_{\mathcal{B}} = \boldsymbol{b} \qquad \text{and} \qquad \boldsymbol{x}_{\mathcal{N}} = 0$$

The basis of the slack variables (1/3)

- Since in $\mathcal{B}$ there are the indices of the m slack variables, we have:

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = \boldsymbol{I} = \boldsymbol{A}_{\mathcal{B}}^{-1} \qquad \text{and} \qquad \boldsymbol{c}_{\mathcal{B}} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

- Accordingly, the matrix $\boldsymbol{A}_{\mathcal{B}}$ is invertible and then it is a basis

The basis of the slack variables (2/3)

- Moreover, we have:

$$\boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad \text{and} \quad \boldsymbol{c}_{\mathcal{N}} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

The basis of the slack variables (3/3)

- In these cases, we then have:

$$\begin{cases} \bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} = \boldsymbol{I}\, \boldsymbol{b} = \boldsymbol{b} \\ \bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}} = \boldsymbol{I}\, \boldsymbol{A}_{\mathcal{N}} = \boldsymbol{A}_{\mathcal{N}} \\ \bar{z} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{I}\, \boldsymbol{b} = 0 \\ \bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}} = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{I}\, \boldsymbol{A}_{\mathcal{N}} = \boldsymbol{c}'_{\mathcal{N}} \end{cases}$$

- The initial dictionary is read directly from the data of the LP problem

Example: initial basic feasible solution (1/2)

- For the example problem, since $\begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, with $\mathcal{B} = \{3, 4, 5\}$ and $\mathcal{N} = \{1, 2\}$, we have the basic feasible solution

$$\begin{pmatrix} x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- Moreover $\bar{z} = 0$

Example: initial basic feasible solution (2/2)

- We have:

$$\boldsymbol{x}_{\mathcal{B}} = \begin{pmatrix} x_3 \\ x_4 \\ x_5 \end{pmatrix}, \quad \boldsymbol{c}'_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$\boldsymbol{x}_{\mathcal{N}} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \underbrace{\boldsymbol{c}'_{\mathcal{N}}}_{=\bar{\boldsymbol{c}}'} = \begin{pmatrix} 5 & 2 \end{pmatrix}, \quad \underbrace{\boldsymbol{A}_{\mathcal{N}}}_{=\bar{\boldsymbol{A}}} = \begin{pmatrix} -1 & 3 \\ 1 & 2 \\ 3 & -2 \end{pmatrix}$$

Summary (1/2)

Bases and dictionaries:

- A **basis** $\boldsymbol{A}_{\mathcal{B}}$ is an invertible matrix of m columns of $\boldsymbol{A}$; it partitions the indices into $\mathcal{B}$ (basic) and $\mathcal{N}$ (non-basic)
- The **dictionary** writes $\boldsymbol{x}_{\mathcal{B}}$ and z in function of $\boldsymbol{x}_{\mathcal{N}}$: $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \bar{\boldsymbol{A}}\,\boldsymbol{x}_{\mathcal{N}}$, $z = \bar{z} + \bar{\boldsymbol{c}}'\,\boldsymbol{x}_{\mathcal{N}}$

Basic solutions:

- Setting $\boldsymbol{x}_{\mathcal{N}} = 0$ gives the **basic solution** $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$
- It is a **basic feasible solution** if $\bar{\boldsymbol{b}} \geq 0$

Summary (2/2)

Reduced costs:

- $\bar{c}_p = c_p - \boldsymbol{c}'_{\mathcal{B}} \boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_p$ is the change of z per unit increment of the non-basic variable x_p

Initial basic feasible solution:

- If $\boldsymbol{b} \geq 0$, the slack variables form the basis $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$
- Then $\bar{\boldsymbol{b}} = \boldsymbol{b}$, $\bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{N}}$, $\bar{z} = 0$ and $\bar{\boldsymbol{c}} = \boldsymbol{c}_{\mathcal{N}}$

Pivoting operations

Introduction: Pivoting operations

- A pivoting operation moves from a basic feasible solution to a **new** one, with a better (or equal) value
- We see how to choose the variable that enters the basis (the **pivot column**) and the one that leaves it (the **pivot row**)
- We see when the LP problem is **unbounded**, with a certificate
- Finally, we see how to update the dictionary: this is the core step of the simplex algorithm

When is a pivoting operation needed?

- Consider a basis $\boldsymbol{A}_{\mathcal{B}}$ and a basic feasible solution $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ of the LP problem in standard form
- If in the associated dictionary there exists $j \in \mathcal{N}$ such that $\bar{c}_j > 0$ (a positive reduced cost), it is necessary to perform a pivoting operation

Definition 7: pivoting operation

*A **pivoting operation** consists in determining a **new basic feasible solution** with the goal of increasing the basic feasible solution value by **swapping one non-basic variable and one basic variable**.*

- *The non-basic variable that becomes basic is called the **pivot column***
- *The basic variable that becomes non-basic is called the **pivot row***

The pivot column

The pivot column is a non-basic variable with a positive reduced cost,

i.e., a non-basic variable

$$x_p \quad \text{with} \quad p \in \mathcal{N} \quad \text{such that} \quad \underbrace{c_p - \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_p}_{=\bar{c}_p} > 0$$

Increasing the pivot column

- Among the non-basic variables, only the value of the pivot column x_p with $p \in \mathcal{N}$ is (potentially) increased, so we have:

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \underbrace{\boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_p}_{=\bar{\boldsymbol{A}}_p} x_p$$

  where $\bar{\boldsymbol{A}}_p$ is the column of the matrix $\bar{\boldsymbol{A}}$ associated to the pivot column x_p

- These m equalities determine the change of the values of the basic variables caused by an increment of the value of the pivot column

The ratio test

- **The value of the pivot column is increased as much as possible while at the same time maintaining non-negative the value of all basic variables**
- In this way a new basic feasible solution is determined
- The value of the pivot column becomes the maximum value t, called t_m, such that:

$$\bar{\boldsymbol{b}} - t \underbrace{\boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_p}_{=\bar{\boldsymbol{A}}_p} \geq 0 \quad \Rightarrow \quad t_m = \min \left\{ \frac{\bar{b}_i}{\bar{a}_{ip}} : i \in \mathcal{B},\ \bar{a}_{ip} > 0 \right\}$$

No positive value in the pivot column

- If $\bar{a}_{ip} \leq 0$ for every $i \in \mathcal{B}$, then the LP problem is unbounded in the direction of x_p
- In this case, the value of the pivot column can be increased without bound while maintaining non-negative the value of all basic variables

Certificate of unboundedness

- Given a basis $\boldsymbol{A}_{\mathcal{B}}$ with a basic feasible solution, if there exists a non-basic variable x_p with $p \in \mathcal{N}$ such that

$$\bar{c}_p > 0 \qquad \text{and} \qquad \bar{a}_{ip} \leq 0, \quad \forall i \in \mathcal{B},$$

  then the LP problem in standard form is **unbounded**

Why the certificate works (1/2)

- Indeed, for every value $t \geq 0$, the point

$$\begin{aligned} x_p &= t, \\ x_i &= \bar{b}_i - t\ \bar{a}_{ip}, \quad \forall i \in \mathcal{B}, \\ x_j &= 0, \quad \forall j \in \mathcal{N} \setminus \{p\}, \end{aligned}$$

  is a feasible solution of the LP problem in standard form

- It satisfies the equality constraints by construction
- It satisfies the non-negativity constraints since $\bar{b}_i \geq 0$ and $-\ t\ \bar{a}_{ip} \geq 0$ for every $i \in \mathcal{B}$

Why the certificate works (2/2)

- The objective value of this feasible solution is

$$\bar{z} + t\ \bar{c}_p$$

- Then, for every value $M \in \mathbb{R}$, the feasible solution obtained with

$$t > \frac{M - \bar{z}}{\bar{c}_p}$$

  has an objective value larger than M

- No value M bounds the objective function from above: the LP problem is unbounded

The pivot row

The pivot row is a basic variable whose value becomes zero by setting the value of the pivot column to t_m,
i.e., a basic variable

$$x_r \quad \text{with} \quad r \in \mathcal{B} \quad \text{such that} \quad \bar{a}_{rp} > 0 \quad \text{and} \quad \bar{b}_r - t_m\, \bar{a}_{rp} = 0$$

$$(\text{i.e.,} \quad \frac{\bar{b}_r}{\bar{a}_{rp}} = t_m)$$

where $\bar{a}_{rp}$ is the value of the column $\bar{\boldsymbol{A}}_p$ associated to the basic variable x_r.

The new basic feasible solution

- Given t_m and $\bar{\boldsymbol{A}}_p$, **the new basic feasible solution** (which determines the new vector $\bar{\boldsymbol{b}}$) is obtained as follows:

$$x_i = \underbrace{\bar{b}_i - t_m\, \bar{a}_{ip}}_{\text{new } \bar{b}_i}, \quad \forall i \in \mathcal{B} \setminus \{r\},$$

$$x_p = \underbrace{t_m}_{\bar{b}_p},$$

$$x_r = 0,$$

$$x_j = 0, \quad \forall j \in \mathcal{N} \setminus \{p\}.$$

The new partition

- Given a pivot column x_p with $p \in \mathcal{N}$ and a pivot row x_r with $r \in \mathcal{B}$, **the new partition of the indices of the variables** is updated as follows:

$$\mathcal{B} \leftarrow \left(\mathcal{B} \setminus \{r\}\right) \cup \{p\} \quad \text{and} \quad \mathcal{N} \leftarrow \left(\mathcal{N} \setminus \{p\}\right) \cup \{r\}$$

- By construction $\boldsymbol{A}_{\mathcal{B}}$ remains a basis

Let us perform a pivoting operation
on the example problem,
starting from the basis of the slack variables.

Example: pivoting operations (1/4)

- For the example problem, with $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$, we have:

$$\boldsymbol{x}_{\mathcal{B}} = \begin{pmatrix} x_3 \\ x_4 \\ x_5 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{c}'_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix} \quad \bar{\boldsymbol{b}} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

$$\boldsymbol{x}_{\mathcal{N}} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \underbrace{\boldsymbol{A}_{\mathcal{N}}}_{=\bar{\boldsymbol{A}}} = \begin{pmatrix} -1 & 3 \\ 1 & 2 \\ 3 & -2 \end{pmatrix} \quad \underbrace{\boldsymbol{c}'_{\mathcal{N}}}_{=\bar{\boldsymbol{c}}'} = \begin{pmatrix} 5 & 2 \end{pmatrix}$$

Example: pivoting operations (2/4)

- We can choose x_1 as the pivot column since $\bar{c}_1 = 5 > 0$
- We then find the largest t such that $\bar{\boldsymbol{b}} - t\,\bar{\boldsymbol{A}}_1 \geq 0$:

$$\begin{cases} \underbrace{12 - t \cdot (-1) \geq 0}_{x_3 \geq 0} \\ \underbrace{12 - t \cdot 1 \geq 0}_{x_4 \geq 0} \\ \underbrace{8 - t \cdot 3 \geq 0}_{x_5 \geq 0} \end{cases} \qquad \begin{cases} t \geq -12 \\ t \leq 12 \\ t \leq \frac{8}{3} \end{cases} \qquad \text{then} \quad t_m = \frac{8}{3}$$

Example: pivoting operations (3/4)

- The value of the pivot column becomes t_m and $x_1 = \frac{8}{3}$
- The value of x_5 becomes zero and x_5 is the pivot row
- The other basic variables take values $\bar{\boldsymbol{b}} - t_m \, \bar{\boldsymbol{A}}_1$, then we have:

$$x_3 = 12 - \frac{8}{3} \cdot (-1) = \frac{44}{3}, \quad x_4 = 12 - \frac{8}{3} \cdot 1 = \frac{28}{3}$$

and accordingly

$$\begin{pmatrix} x_1 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} \frac{8}{3} \\ \frac{44}{3} \\ \frac{28}{3} \end{pmatrix}$$

- The new partition of the indices of the variables is $\mathcal{B} = \{1, 3, 4\}$ and $\mathcal{N} = \{2, 5\}$

Example: pivoting operations (4/4)

- In the same dictionary also $\bar{c}_2 = 2 > 0$, so x_2 can be chosen as the pivot column as well
- The ratio test then gives $t_m = 4$, the pivot row is x_3 and the new partition is $\mathcal{B} = \{2, 4, 5\}$ and $\mathcal{N} = \{1, 3\}$
- The execution of the simplex algorithm that starts with this choice is worked out in the exercises of this chapter (*Simplex algorithm with a different pivot column*)

Now an example in which
no pivot row can be found:

the certificate of unboundedness at work.

Example: certificate of unboundedness (1/5)

- Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 2$ less-than-or-equal-to linear constraints:

$$\begin{array}{rrrrrrr} \max & & x_1 & + & x_2 & & \\ \text{subject to} & - & x_1 & + & x_2 & \leq & 1 \\ & & x_1 & - & x_2 & \leq & 1 \\ & & x_1, & & x_2 & \geq & 0 \end{array}$$

Example: certificate of unboundedness (2/5)

- By adding $m = 2$ slack variables, with $\mathcal{B} = \{3, \ 4\}$ and $\mathcal{N} = \{1, \ 2\}$ we obtain the initial dictionary:

$$\begin{cases} x_3 = 1 + x_1 - x_2 \\ x_4 = 1 - x_1 + x_2 \\ z = 0 + x_1 + x_2 \end{cases}$$

- We perform one pivoting operation with pivot column x_1 (since $\bar{c}_1 = 1 > 0$)
- The only positive value in the column of x_1 is $\bar{a}_{41} = 1$, and then $t_m = \frac{1}{1} = 1$ and the pivot row is x_4

Example: certificate of unboundedness (3/5)

- With $\mathcal{B} = \{1, \ 3\}$ and $\mathcal{N} = \{2, \ 4\}$, the new dictionary is:

$$\begin{cases} x_1 = 1 + x_2 - x_4 \\ x_3 = 2 - x_4 \\ z = 1 + 2\,x_2 - x_4 \end{cases}$$

- The reduced cost of the non-basic variable x_2 is $\bar{c}_2 = 2 > 0$, and the values of its column are

$$\bar{a}_{12} = -1 \leq 0 \qquad \text{and} \qquad \bar{a}_{32} = 0 \leq 0.$$

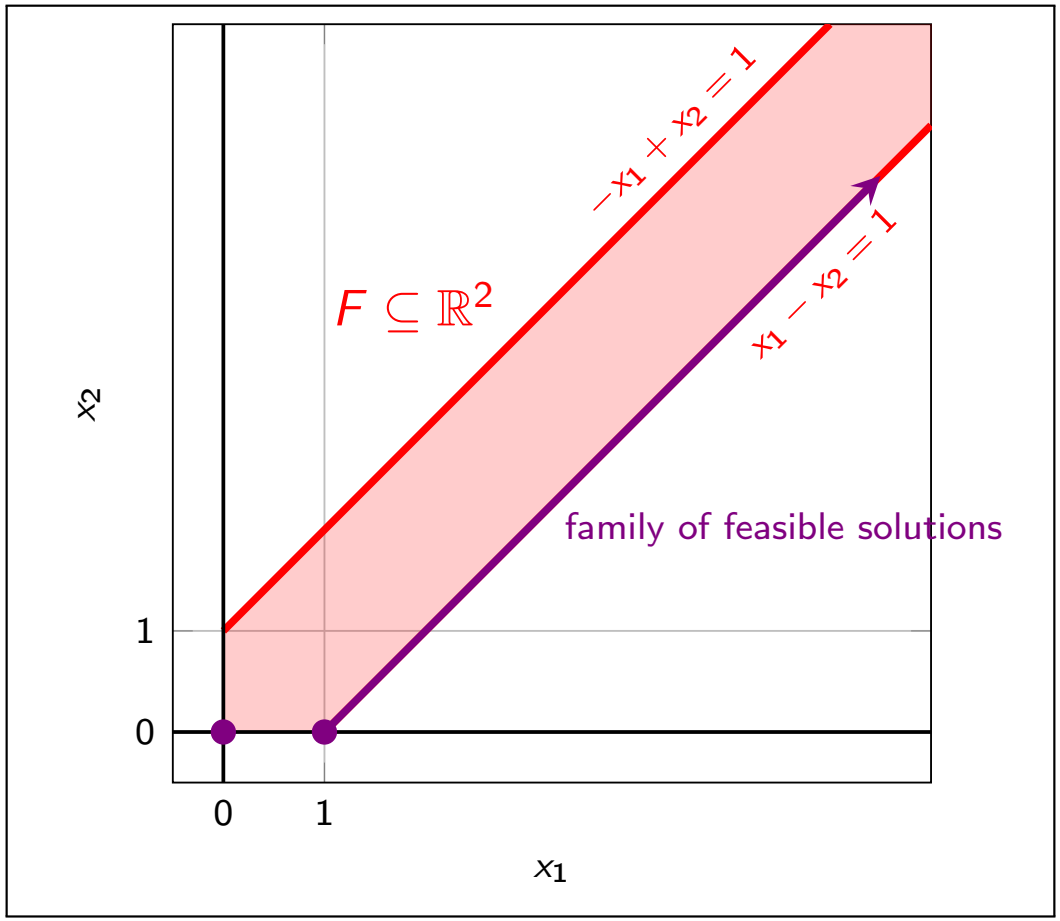
Example: certificate of unboundedness (4/5)

- Accordingly, no pivot row can be determined and this LP problem is **unbounded**
- The corresponding family of feasible solutions is, for every value $t \geq 0$:

$$x_2 = t, \qquad x_1 = 1 + t, \qquad x_3 = 2, \qquad x_4 = 0,$$

whose objective value is $1 + 2\,t$

Example: certificate of unboundedness (5/5)



In violet the basic feasible solution reached after the first pivoting operation and the family of feasible solutions along which the LP problem is unbounded.

Update of the dictionary

Once the pivot column x_p with $p \in \mathcal{N}$
and the pivot row x_r with $r \in \mathcal{B}$ are determined,

the new dictionary can be determined
by appropriate row operations to achieve the interchange.

The equation of the pivot column

- Let x_p be the pivot column and x_r the pivot row. From the equation of the pivot row, we have:

$$x_r = \bar{b}_r - \bar{a}_{rp} x_p - \sum_{j \in \mathcal{N} \setminus \{p\}} \bar{a}_{rj} x_j.$$

- Solving for x_p gives the equation of the pivot column in the new dictionary:

$$x_p = \frac{\bar{b}_r}{\bar{a}_{rp}} - \sum_{j \in \mathcal{N} \setminus \{p\}} \frac{\bar{a}_{rj}}{\bar{a}_{rp}} x_j - \frac{1}{\bar{a}_{rp}} x_r.$$

The equations of the other basic variables

- For each remaining basic variable $i \in \mathcal{B} \setminus \{r\}$, substituting the equation of the pivot column gives their equations in the new dictionary:

$$x_i = \bar{b}_i - \sum_{j \in \mathcal{N} \setminus \{p\}} \bar{a}_{ij} x_j - \bar{a}_{ip} \left(\frac{\bar{b}_r}{\bar{a}_{rp}} - \sum_{j \in \mathcal{N} \setminus \{p\}} \frac{\bar{a}_{rj}}{\bar{a}_{rp}} x_j - \frac{1}{\bar{a}_{rp}} x_r \right).$$

The equation of the objective function

- For the objective function, substituting the equation of the pivot column gives the equation of the objective function in the new dictionary:

$$z = \bar{z} + \sum_{j \in \mathcal{N} \setminus \{p\}} \bar{c}_j x_j + \bar{c}_p \left(\frac{\bar{b}_r}{\bar{a}_{rp}} - \sum_{j \in \mathcal{N} \setminus \{p\}} \frac{\bar{a}_{rj}}{\bar{a}_{rp}} x_j - \frac{1}{\bar{a}_{rp}} x_r \right).$$

Example: update of the dictionary (1/4)

- For the example problem, with $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$, we have the dictionary:

$$\begin{cases} x_3 = 12 + x_1 - 3\,x_2 \\ x_4 = 12 - x_1 - 2\,x_2 \\ x_5 = 8 - 3\,x_1 + 2\,x_2 \\ z = 0 + 5\,x_1 + 2\,x_2 \end{cases}$$

Example: update of the dictionary (2/4)

- Performing the pivot operation with x_1 as the pivot column and x_5 as the pivot row, we have:

$$x_5 = 8 - 3\,x_1 + 2\,x_2 \quad \text{then} \quad x_1 = \tfrac{8}{3} + \tfrac{2}{3}\,x_2 - \tfrac{1}{3}\,x_5$$

- Substituting in the other equations of the dictionary, we get:

$$x_3 = 12 + x_1 - 3\,x_2 = 12 + \left(\frac{8}{3} + \frac{2}{3}\,x_2 - \frac{1}{3}\,x_5\right) - 3\,x_2$$

$$= \frac{44}{3} - \frac{7}{3}\,x_2 - \frac{1}{3}\,x_5$$

Example: update of the dictionary (3/4)

$$x_4 = 12 - x_1 - 2\,x_2 = 12 - \left(\frac{8}{3} + \frac{2}{3}\,x_2 - \frac{1}{3}\,x_5\right) - 2\,x_2$$

$$= \frac{28}{3} - \frac{8}{3}\,x_2 + \frac{1}{3}\,x_5$$

$$z = 5\,x_1 + 2\,x_2 = 5\left(\frac{8}{3} + \frac{2}{3}\,x_2 - \frac{1}{3}\,x_5\right) + 2\,x_2$$

$$= \frac{40}{3} + \frac{16}{3}\,x_2 - \frac{5}{3}\,x_5$$

Example: update of the dictionary (4/4)

- Accordingly the new dictionary reads as follows:

$$\begin{cases} x_1 = \frac{8}{3} + \frac{2}{3} x_2 - \frac{1}{3} x_5 \\ x_3 = \frac{44}{3} - \frac{7}{3} x_2 - \frac{1}{3} x_5 \\ x_4 = \frac{28}{3} - \frac{8}{3} x_2 + \frac{1}{3} x_5 \\ z = \frac{40}{3} + \frac{16}{3} x_2 - \frac{5}{3} x_5 \end{cases}$$

- It is associated to the feasible solution $\bar{x}_1 = \frac{8}{3},\ \bar{x}_2 = 0,\ \bar{x}_3 = \frac{44}{3},\ \bar{x}_4 = \frac{28}{3},\ \bar{x}_5 = 0$ of value $z = \frac{40}{3}$: the same dictionary computed above with $\mathcal{B} = \{1, 3, 4\}$

Summary (1/2)

Pivoting operation:

- It swaps a non-basic variable (the **pivot column**) and a basic variable (the **pivot row**), to increase the value of the basic feasible solution
- Pivot column: a non-basic variable x_p with $\bar{c}_p > 0$
- Ratio test: $t_m = \min\{\bar{b}_i/\bar{a}_{ip} : i \in \mathcal{B},\ \bar{a}_{ip} > 0\}$
- Pivot row: a basic variable x_r with $\bar{a}_{rp} > 0$ and $\bar{b}_r/\bar{a}_{rp} = t_m$

Unboundedness:

- If $\bar{c}_p > 0$ and $\bar{a}_{ip} \leq 0$ for all $i \in \mathcal{B}$, the LP problem is unbounded: the family $x_p = t$, $x_i = \bar{b}_i - t\,\bar{a}_{ip}$ has value $\bar{z} + t\,\bar{c}_p$

Summary (2/2)

Update of the dictionary:

- Isolate x_p in the equation of the pivot row
- Substitute it in the equations of the other basic variables and of z
- Update the partition: $\mathcal{B} \leftarrow (\mathcal{B} \setminus \{r\}) \cup \{p\}$ and $\mathcal{N} \leftarrow (\mathcal{N} \setminus \{p\}) \cup \{r\}$

In the example problem:

- From $\mathcal{B} = \{3, 4, 5\}$, with pivot column x_1 and pivot row x_5, we reach $\mathcal{B} = \{1, 3, 4\}$ of value $\frac{40}{3}$

High-level pseudo-code of the simplex algorithm

Introduction: High-level pseudo-code of the simplex algorithm

- We put all the pieces together: initialization, optimality check, pivoting operations
- The pseudo-code also says what to do when no initial basic feasible solution is at hand (Phase 0)
- We then execute the simplex algorithm on the example problem, iteration by iteration, and follow its path on the polyhedron

Phase 0 of the simplex algorithm

- The **Phase 0 of the simplex algorithm** consists in a procedure to determine an initial basic feasible solution and the associated initial dictionary
- The Phase 0 is also able to prove that no basic feasible solution exists for an LP problem in standard form

Simplex algorithm (1/9)

Input

An LP problem in standard form.

Output

An optimal solution of the LP problem, if one exists; otherwise, a certificate that the LP problem is infeasible or unbounded.

Step 0: **Initialization**

If $b_i \geq 0$ for all $i \in \{1, 2, \ldots, m\}$, then set

$$\mathcal{B} \leftarrow \{n+1, n+2, \ldots, n+m\},$$
$$\mathcal{N} \leftarrow \{1, 2, \ldots, n\},$$

Simplex algorithm (2/9)

Step 0: **Initialization** (continued)
and consider the basic feasible solution

$$x_i = \bar{b}_i, \quad \forall i \in \mathcal{B}, \qquad x_j = 0, \quad \forall j \in \mathcal{N},$$

of the LP problem, together with the associated initial dictionary. Otherwise, execute Phase 0 of the simplex algorithm.
If Phase 0 proves that no basic feasible solution exists, then the LP problem is **infeasible** and `STOP`. Otherwise, consider the basic feasible solution found by Phase 0 together with its dictionary.

Simplex algorithm (3/9)

Step 1: **Optimality check**

With respect to the current dictionary

$$\begin{cases} x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij} x_j, & \forall i \in \mathcal{B}, \\ z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j x_j, \end{cases}$$

check the reduced costs.

Simplex algorithm (4/9)

Step 1: **Optimality check** (continued)

If

$$\bar{c}_j \leq 0, \qquad \forall j \in \mathcal{N},$$

then the basic feasible solution

$$x_i = \bar{b}_i, \quad \forall i \in \mathcal{B}, \qquad x_j = 0, \quad \forall j \in \mathcal{N},$$

is an optimal solution and $\bar{z}$ is the optimal value of the LP problem. In this case `STOP`.

Simplex algorithm (5/9)

Step 2: **Pivoting operations**

The **pivot column** is a non-basic variable x_p with $p \in \mathcal{N}$ such that:

$$\bar{c}_p > 0$$

a. If

$$\bar{a}_{ip} \leq 0, \quad \forall i \in \mathcal{B},$$

then the LP problem is **unbounded**. `STOP`.

b. Find the largest value t (called t_m) such that:

$$\bar{b}_i - t\ \bar{a}_{ip} \geq 0, \quad \forall i \in \mathcal{B}$$

$$\Rightarrow \quad t_m \leftarrow \min \left\{ \frac{\bar{b}_i}{\bar{a}_{ip}} : i \in \mathcal{B},\ \bar{a}_{ip} > 0 \right\}$$

Simplex algorithm (6/9)

Step 2: **Pivoting operations** (continued)

The **pivot row** is a basic variable x_r with $r \in \mathcal{B}$ such that:

$$\bar{a}_{rp} > 0 \quad \text{and} \quad \bar{b}_r - t_m \, \bar{a}_{rp} = 0$$

c. Update the dictionary with the new basis by performing the following operations:
   - The new equation of the pivot column x_p is obtained by isolating x_p in the pivot row:

$$x_p = \frac{\bar{b}_r}{\bar{a}_{rp}} - \sum_{j \in \mathcal{N} \setminus \{p\}} \frac{\bar{a}_{rj}}{\bar{a}_{rp}} x_j - \frac{1}{\bar{a}_{rp}} x_r.$$

Simplex algorithm (7/9)

Step 2: **Pivoting operations** (continued)

c. (continued)

- The new equations of the basic variables x_i with $i \in \mathcal{B} \setminus \{r\}$ are obtained by substituting the new equation of x_p in the old equations of x_i:

$$x_i = \bar{b}_i - \sum_{j \in \mathcal{N} \setminus \{p\}} \bar{a}_{ij} x_j - \bar{a}_{ip} \left(\frac{\bar{b}_r}{\bar{a}_{rp}} - \sum_{j \in \mathcal{N} \setminus \{p\}} \frac{\bar{a}_{rj}}{\bar{a}_{rp}} x_j - \frac{1}{\bar{a}_{rp}} x_r \right).$$

Simplex algorithm (8/9)

Step 2: **Pivoting operations** (continued)

c. (continued)

- The new equation of the objective function z is obtained by substituting the new equation of x_p in the old equation of z:

$$z = \bar{z} + \sum_{j \in \mathcal{N} \setminus \{p\}} \bar{c}_j x_j + \bar{c}_p \left(\frac{\bar{b}_r}{\bar{a}_{rp}} - \sum_{j \in \mathcal{N} \setminus \{p\}} \frac{\bar{a}_{rj}}{\bar{a}_{rp}} x_j - \frac{1}{\bar{a}_{rp}} x_r \right).$$

Simplex algorithm (9/9)

Step 2: **Pivoting operations** (continued)

d. Update the basis with the new partition of the indices of the variables:

$$\mathcal{B} \leftarrow (\mathcal{B} \setminus \{r\}) \cup \{p\}$$
$$\mathcal{N} \leftarrow (\mathcal{N} \setminus \{p\}) \cup \{r\}$$

go to step 1.

We now execute the simplex algorithm
on the example problem,
one iteration at a time.

Example: execution of the simplex algorithm

- We execute the simplex algorithm on the example problem
- Since $\boldsymbol{b} \geq 0$, the initial basic feasible solution is the one of the example on the initial basic feasible solution, associated with $\mathcal{B} = \{3, 4, 5\}$ and $\mathcal{N} = \{1, 2\}$

Example: initialization

- Consider the initial dictionary:

$$\begin{cases} x_3 = 12 + x_1 - 3x_2 \\ x_4 = 12 - x_1 - 2x_2 \\ x_5 = 8 - 3x_1 + 2x_2 \\ z = 0 + 5x_1 + 2x_2 \end{cases}$$

- $\mathcal{B} = \{3, 4, 5\}$ and $\mathcal{N} = \{1, 2\}$. The basic feasible solution is:

$$\bar{x}_3 = 12, \quad \bar{x}_4 = 12, \quad \bar{x}_5 = 8, \quad \bar{x}_1 = 0, \quad \bar{x}_2 = 0 \quad \text{of value} \quad 0$$

Example: iteration 1 (1/3)

- If we increase the value of the variable x_1, we can get a better solution. We have:

$$\begin{cases} x_3 = 12 + x_1 \\ x_4 = 12 - x_1 \\ x_5 = 8 - 3\,x_1 \end{cases} \qquad \begin{cases} \underbrace{12 + x_1 \geq 0}_{x_3 \geq 0} \\ \underbrace{12 - x_1 \geq 0}_{x_4 \geq 0} \\ \underbrace{8 - 3\,x_1 \geq 0}_{x_5 \geq 0} \end{cases} \qquad \begin{cases} x_1 \geq -12 \\ x_1 \leq 12 \\ x_1 \leq \dfrac{8}{3} \end{cases}$$

- By increasing the value of x_1 to $\min\left\{12, \frac{8}{3}\right\} = \frac{8}{3}$, the value of x_5 decreases to 0

Example: iteration 1 (2/3)

- Then, we have that: x_1 becomes basic and accordingly x_5 becomes non-basic
- We have:

$$x_5 = 8 - 3\,x_1 + 2\,x_2 \quad \text{then} \quad x_1 = \frac{8}{3} + \frac{2}{3}\,x_2 - \frac{1}{3}\,x_5.$$

- Substituting in the other equations of the dictionary, we get:

$$x_3 = 12 + \left(\frac{8}{3} + \frac{2}{3}\,x_2 - \frac{1}{3}\,x_5\right) - 3\,x_2 = \frac{44}{3} - \frac{7}{3}\,x_2 - \frac{1}{3}\,x_5$$

Example: iteration 1 (3/3)

$$x_4 = 12 - \left(\frac{8}{3} + \frac{2}{3}\,x_2 - \frac{1}{3}\,x_5\right) - 2\,x_2 = \frac{28}{3} - \frac{8}{3}\,x_2 + \frac{1}{3}\,x_5$$

$$z = 5\left(\frac{8}{3} + \frac{2}{3}\,x_2 - \frac{1}{3}\,x_5\right) + 2\,x_2 = \frac{40}{3} + \frac{16}{3}\,x_2 - \frac{5}{3}\,x_5$$

Example: iteration 2 (1/5)

- Consider the dictionary:

$$\begin{cases} x_1 = \frac{8}{3} + \frac{2}{3}x_2 - \frac{1}{3}x_5 \\ x_3 = \frac{44}{3} - \frac{7}{3}x_2 - \frac{1}{3}x_5 \\ x_4 = \frac{28}{3} - \frac{8}{3}x_2 + \frac{1}{3}x_5 \\ z = \frac{40}{3} + \frac{16}{3}x_2 - \frac{5}{3}x_5 \end{cases}$$

- $\mathcal{B} = \{1, 3, 4\}$ and $\mathcal{N} = \{2, 5\}$. The basic feasible solution is:

$$\bar{x}_1 = \frac{8}{3}, \quad \bar{x}_3 = \frac{44}{3}, \quad \bar{x}_4 = \frac{28}{3}, \quad \bar{x}_2 = 0, \quad \bar{x}_5 = 0 \quad \text{of value} \quad \frac{40}{3}$$

Example: iteration 2 (2/5)

- If we increase the value of the variable x_2, we can get a better solution. We have:

$$\begin{cases} x_1 = \frac{8}{3} + \frac{2}{3} x_2 \\ x_3 = \frac{44}{3} - \frac{7}{3} x_2 \\ x_4 = \frac{28}{3} - \frac{8}{3} x_2 \end{cases} \qquad \begin{cases} \overbrace{\frac{8}{3} + \frac{2}{3} x_2 \geq 0}^{x_1 \geq 0} \\ \underbrace{\frac{44}{3} - \frac{7}{3} x_2 \geq 0}_{x_3 \geq 0} \\ \underbrace{\frac{28}{3} - \frac{8}{3} x_2 \geq 0}_{x_4 \geq 0} \end{cases} \qquad \begin{cases} x_2 \geq -4 \\ x_2 \leq \frac{44}{7} \\ x_2 \leq \frac{7}{2} \end{cases}$$

- By increasing the value of x_2 to $\min\left\{\frac{44}{7}, \frac{7}{2}\right\} = \frac{7}{2}$, the value of x_4 decreases to 0

Example: iteration 2 (3/5)

- Then, we have that: x_2 becomes basic and accordingly x_4 becomes non-basic

- We have:

$$x_4 = \frac{28}{3} - \frac{8}{3}\,x_2 + \frac{1}{3}\,x_5 \quad \text{then} \quad x_2 = \frac{7}{2} - \frac{3}{8}\,x_4 + \frac{1}{8}\,x_5.$$

- Substituting in the other equations of the dictionary, we get:

$$\begin{aligned} x_1 &= \frac{8}{3} + \frac{2}{3}\left(\frac{7}{2} - \frac{3}{8}\,x_4 + \frac{1}{8}\,x_5\right) - \frac{1}{3}\,x_5 \\ &= 5 - \frac{1}{4}\,x_4 - \frac{1}{4}\,x_5 \end{aligned}$$

Example: iteration 2 (4/5)

$$x_3 = \frac{44}{3} - \frac{7}{3}\left(\frac{7}{2} - \frac{3}{8}x_4 + \frac{1}{8}x_5\right) - \frac{1}{3}x_5$$

$$= \frac{13}{2} + \frac{7}{8}x_4 - \frac{5}{8}x_5$$

$$z = \frac{40}{3} + \frac{16}{3}\left(\frac{7}{2} - \frac{3}{8}x_4 + \frac{1}{8}x_5\right) - \frac{5}{3}x_5$$

$$= 32 - 2x_4 - x_5$$

Example: iteration 2 (5/5)

- Accordingly, we obtained an optimal solution and this is the optimal dictionary:

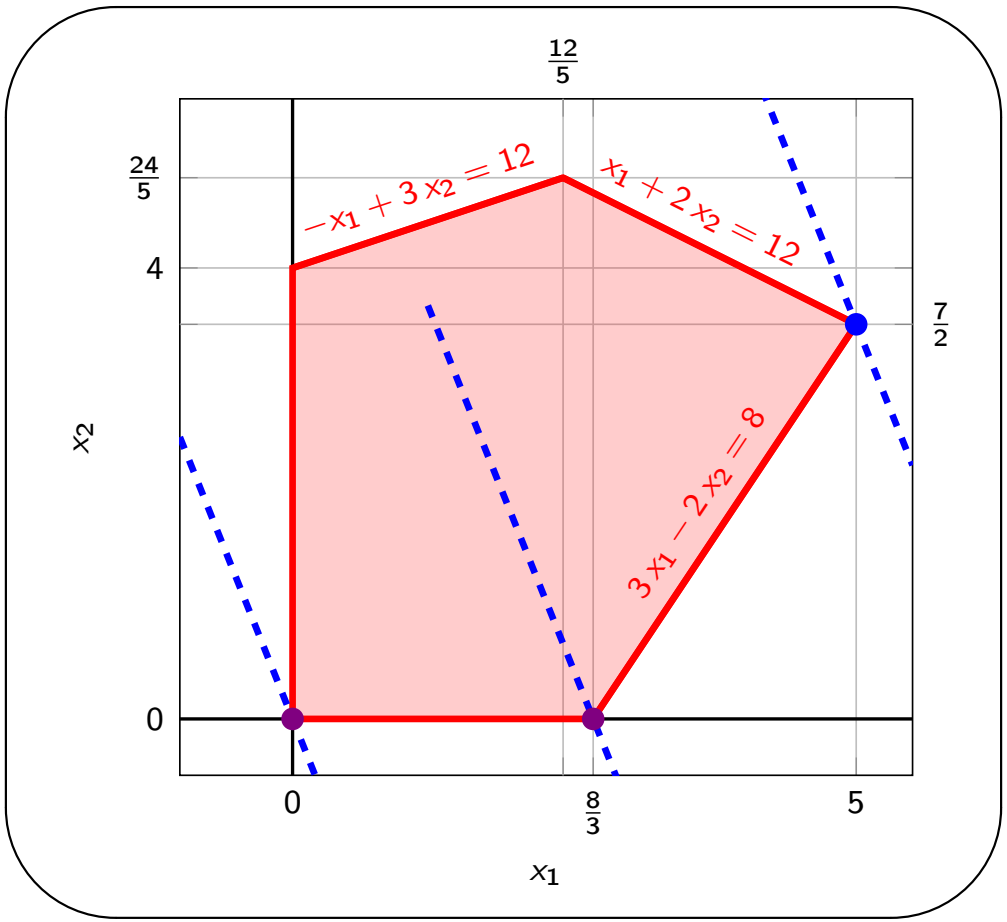
$$\begin{cases} x_1 = 5 - \frac{1}{4} x_4 - \frac{1}{4} x_5 \\ x_2 = \frac{7}{2} - \frac{3}{8} x_4 + \frac{1}{8} x_5 \\ x_3 = \frac{13}{2} + \frac{7}{8} x_4 - \frac{5}{8} x_5 \\ z = 32 - 2 x_4 - x_5 \end{cases}$$

- $\mathcal{B} = \{1, 2, 3\}$ and $\mathcal{N} = \{4, 5\}$. The optimal solution is:

$$\tilde{x}_1 = 5, \quad \tilde{x}_2 = \frac{7}{2}, \quad \tilde{x}_3 = \frac{13}{2}, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0 \quad \text{of value} \quad 32$$

Example: the path on the polyhedron (1/2)

The feasible region of the LP problem (in canonical form) is the polyhedron shaded in red:



Example: the path on the polyhedron (2/2)

- In violet the first and the second vertex explored by this execution of the simplex algorithm, $(0, 0)$ and $(\frac{8}{3}, 0)$, before reaching the optimal one depicted in blue, $(5, \frac{7}{2})$
- In dashed blue, the contour lines

$$5\,x_1 + 2\,x_2 = 0, \quad 5\,x_1 + 2\,x_2 = \frac{40}{3} \quad \text{and} \quad 5\,x_1 + 2\,x_2 = 32$$

of the objective function associated to the values taken in the three vertices

The choice of the pivot column

- At the first iteration the pivot column x_1 was chosen among the two non-basic variables with a positive reduced cost
- Choosing x_2 instead, the simplex algorithm reaches the same optimal solution after three pivoting operations, visiting the vertices $(0, 4)$ and $\left(\frac{12}{5}, \frac{24}{5}\right)$
- The choice of the pivot column changes the path on the polyhedron and the number of iterations (see the exercise *Simplex algorithm with a different pivot column*)

Summary

- **Step 0**: if $\boldsymbol{b} \geq 0$ start from the basis of the slack variables, otherwise run Phase 0 (which may prove that the LP problem is infeasible)
- **Step 1**: if $\bar{c}_j \leq 0$ for all $j \in \mathcal{N}$, the current basic feasible solution is optimal
- **Step 2**: choose a pivot column with $\bar{c}_p > 0$; stop if the problem is unbounded; otherwise ratio test, pivot row, update of the dictionary and of the partition
- On the example problem: two iterations, along the vertices $(0, 0)$, $(\frac{8}{3}, 0)$, $(5, \frac{7}{2})$, optimal value 32

Summary (1/2)

From the problem to the dictionary:

- Standard form with m slack variables: $\boldsymbol{A}$ contains $\boldsymbol{I}$ and $\text{rank}(\boldsymbol{A}) = m$
- A basis $\boldsymbol{A}_{\mathcal{B}}$ gives the dictionary $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \bar{\boldsymbol{A}}\,\boldsymbol{x}_{\mathcal{N}}$, $z = \bar{z} + \bar{\boldsymbol{c}}'\,\boldsymbol{x}_{\mathcal{N}}$
- Basic feasible solution: $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \geq 0$, $\boldsymbol{x}_{\mathcal{N}} = 0$

Reduced costs:

- $\bar{\boldsymbol{c}}$ measures the change of z per unit increment of each non-basic variable

Summary (2/2)

The simplex algorithm:

- Start from a basic feasible solution (slack variables if $\boldsymbol{b} \geq 0$, otherwise Phase 0)
- Stop when all the reduced costs are ≤ 0: the solution is optimal
- Otherwise pivot: pivot column with $\bar{c}_p > 0$, ratio test, pivot row, update of the dictionary
- If the pivot column has no positive value, the LP problem is unbounded

The choice of the pivot column:

- It changes the path on the polyhedron and the number of iterations, not the optimal value